Skills Summary

Function Notation: When $f(x)$ Is Not Multiplication

Use this reference while completing your worksheet or when studying.

Skill 1 — Evaluating Function Notation

What the notation says: $f(\text{input}) = \text{output}$.

f names a *rule*, not a number, so the parentheses hold an input slot — never a factor.

- $f(3)$: replace every x in the rule with 3, then simplify.
- Nothing may be distributed across f: f has no value to distribute.
- If an x survives after you were asked for $f(3)$, you never substituted.

Example: $f(x) = 2x + 7$. Find $f(5)$.
Step 1. The 5 sits in the input slot, so the only legal move is substitution — replace x by 5.
Step 2. $f(5) = 2(5) + 7 = 10 + 7$, so $f(5) = \mathbf{17}$.

Try it: $g(x) = x^2 - 3$. Find $g(4)$.

check: **13**

Skill 2 — Fix a Notation-as-Multiplication Error

Three signatures of the multiplication misreading:

- The work still shows an x after a numeric input was given.
- The whole rule got multiplied by the input, so the answer is input-times-too-big.
- The rule was "distributed" over a sum in the input slot: $f(a + b)$ became $f(a) + f(b)$.

Also keep straight: $2f(3)$ doubles the *output*; $f(2 \cdot 3)$ doubles the *input*.

Example: For $f(x) = 3x - 2$ a student writes $f(4) = 4(3x - 2) = 40$. Fix it.
Step 1. An x is still present and the input was used as a multiplier, so this is the times-the-rule misreading, not an arithmetic slip.
Step 2. Substitute instead: $f(4) = 3(4) - 2 = 12 - 2$, so $f(4) = \mathbf{10}$.

Try it: For $g(x) = x + 6$ a student writes $g(2) = 2(x + 6) = 16$. Fix it.

check: **8**

Skill 3 — Solving for the Input

Two different questions:

- "Find $f(4)$" gives the input, asks for the output: substitute.
- "Find x with $f(x) = 13$" gives the output, asks for the input: set the rule equal to the output and solve.

You can never divide by f to undo it — f is a name, not a factor.

Example: $f(x) = 3x + 1$. Find every x with $f(x) = 13$.
Step 1. The output is known and the input is missing, so replace the output by the rule and solve the equation.
Step 2. $3x + 1 = 13 \Rightarrow 3x = 12 \Rightarrow \boldsymbol{x = 4}$; check $f(4) = 13$ ✓.

Try it: $f(x) = 2x - 5$. Find every x with $f(x) = 9$.

check: $\boldsymbol{x = 7}$